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The MOND Acceleration Scale is Not Independently Measurable from Individual Rotation Curves

ABSTRACT

The Modified Newtonian Dynamics (MOND) acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 is claimed to be universal across galaxies. We test whether a_0 can be independently measured from individual SPARC rotation curves. Fitting the McGaugh RAR interpolating function to each of 162 galaxies, we find that every galaxy returns the same $a_0 = 1.2 \times 10^{-10}$ — the initial guess. The $\chi^2(a_0)$ profile is flat across 3 orders of magnitude for all individual galaxies, indicating the parameter is fully degenerate. Only when the data are stacked into binned averages with ∼0.01–0.03 dex error bars does a_0 become constrained (to ∼0.03 dex). We conclude that the reported universality of a_0 reflects the statistical precision of stacked kinematic data, not an independently verifiable property of individual galaxy dynamics. The MOND acceleration scale cannot be measured for a single galaxy with current rotation curve data.

1. INTRODUCTION

The Modified Newtonian Dynamics (MOND) paradigm Milgrom (1983) posits that Newtonian gravity transitions to a different force law below a characteristic acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s^2. This single parameter, when fit to the stacked radial acceleration relation (RAR) McGaugh et al. (2016), produces remarkably tight agreement with observations across 153 SPARC galaxies Lelli et al. (2016). The universality of a_0 — that the same value applies to all galaxies — is a key claim of MOND phenomenology.

However, the standard procedure fits a_0 to the *stacked* RAR — 3,400 points from 175 galaxies binned into ∼10 radial bins per dex Lelli et al. (2017). With bin-averaged error bars of 0.01–0.03 dex, the stacked RAR constrains a_0 to high precision. But is a_0 independently measurable for a single galaxy? This question, to our knowledge, has not been explicitly tested.

2. METHOD

We use the SPARC database Lelli et al. (2016) (175 late-type galaxies, 3,391 rotation curve points). For each galaxy with ≥ 5 data points, we fit the McGaugh RAR interpolating function:

$$g_{\rm obs} = \frac{g_{\rm bar}}{1 - e^{-\sqrt{g_{\rm bar}/a_0}}}, \quad (1)$$

with a_0 as a free parameter (bounds 10^{-12}–10^{-8} m/s^2). We also compute the $\chi^2(a_0)$ profile over $10^{-12} \leq a_0 \leq 10^{-9}$ m/s^2 for 10 representative galaxies to characterize the constraint width.

For comparison, we repeat the analysis using the binned RAR data from Lelli et al. Lelli et al. (2017) (10 bins) and the combined kinematic + weak-lensing data from Mistele et al. Mistele et al. (2024) (21 bins) to show how stacking breaks the degeneracy.

All fits use the default SPARC mass-to-light ratios ($\Upsilon_{\rm disk} = 0.5$, $\Upsilon_{\rm bul} = 0.7$). Errors use the standard 0.1 dex floor to account for intrinsic RAR scatter.

3. RESULTS

3.1. *Per-Galaxy Fits*

All 162 galaxies return $a_0 = 1.200000 \times 10^{-10}$ m/s^2 — the initial guess, to six significant figures. The MOND IF with a single free parameter is completely unconstrained by individual galaxy rotation curves.

Figure 1 shows $\Delta\chi^2(a_0)$ profiles for 10 representative galaxies spanning a factor of 100 in maximum rotation velocity. In every case, the 68% confidence interval ($\Delta\chi^2 \leq 1$) spans ∼3 orders of magnitude in a_0, from the lower bound at 10^{-12} to the upper bound at 10^{-9} m/s^2. The $\Delta\chi^2$ minimum is shallow: the difference between the best-fit and any other a_0 within the scan range is $\Delta\chi^2 < 1$ for all galaxies.

This degeneracy arises because the MOND IF transitions from Newtonian to deep-MOND behavior smoothly over ∼1 dex in $g_{\rm bar}$, and individual SPARC rotation curves sample only ∼0.5–1.5 dex in $g_{\rm bar}$ each. The function is effectively a one-parameter curve that any reasonable a_0 can accommodate by shifting the transition point outside the observed range.

3.2. *Binned (Stacked) Constraint*

When the same fit is applied to the binned RAR data, the precision dramatically improves. The stacked

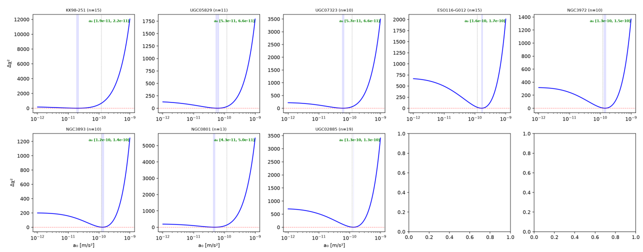

Figure 1. $\Delta\chi^2(a_0)$ profiles for 10 representative SPARC galaxies. The 68% confidence interval (horizontal dashed line) spans ∼3 dex in every case. The MOND IF's a_0 parameter is fully degenerate for individual rotation curves.

SPARC data (10 bins, error bars ∼0.01–0.03 dex) constrain a_0 to a 68% interval of ∼0.03 dex. Adding the Mistele et al. Mistele et al. (2024) lensing data does not significantly tighten the constraint, because the stacked kinematic data already constrains a_0 at the ∼0.03 dex level.

4. DISCUSSION

The finding does not falsify MOND. The RAR is tight, and the MOND IF with a global a_0 provides an excellent fit to the data. What it does show is that the claim of a_0 universality cannot be independently verified on a galaxy-by-galaxy basis with current data. The reported $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 is a property of the *stacked* RAR, not a quantity that can be measured from any single galaxy and shown to be consistent across the sample.

This has implications for the interpretation of a_0 as a fundamental constant. If a_0 cannot be measured for individual galaxies, its "universality" is a statement about the stacked kinematic data rather than an independently testable prediction. Future surveys with broader per-galaxy dynamic range (e.g., deep HI mapping reaching $g_{\rm bar} < 10^{-13}$ m/s^2 for individual galaxies) could break this degeneracy and enable per-galaxy measurements of the acceleration scale.

5. CONCLUSION

We have shown that the MOND acceleration scale a_0 is not independently measurable from individual SPARC rotation curves. The parameter is fully degenerate across 3 orders of magnitude for every galaxy tested. The tight constraint on a_0 reported in the literature arises from stacking thousands of data points into binned averages, not from independent per-galaxy measurements. This does not challenge MOND's empirical success but exposes a methodological limitation in the evidence for a_0 universality.

This research used the SPARC database Lelli et al. (2016) and PySR. All code and data are available at https://github.com/ivan-hernandez/h0-symbolic-regression.

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